

Molecular dynamic simulations of molecules at Teflon surface: towards understanding antibody adsorption in the microcapillary film (MCF) rapid diagnostic platform

Alice Picas, Ana Isabel Barbosa, Carmelo Moreno and Nuno M. Reis

Center of Biological Engineering, University of Minho, 4710-057 Braga, Portugal

Abstract. Coarse-grained molecular dynamics study of molecules in Teflon surfaces for mechanistic understanding of antibody adsorption onto teflon microcapillaries, towards design and performance improvement of miniaturized, microfluidic, quantitative rapid diagnostics.

Keywords: Microfluidic diagnostics; FEP-Teflon; Antibody adsorption; Molecular Dynamics; Computer simulations

1 Introduction and State-of-the-Art

Point-of-care (POC) diagnostics enable decentralized testing, yet most current technologies remain qualitative. Quantitative capabilities are essential for disease monitoring and early diagnosis. Protein biomarkers, whose concentrations in biological fluids are closely linked to these conditions, are therefore critical targets, making their accurate quantification key for diagnosis, prognosis, and monitoring of therapeutic response [1], [2], [3], [4], [5].

Microfluidic diagnostics have emerged as a key technology for rapid quantification in POC applications due to their ability to precisely manipulate small fluid volumes, reducing reagent consumption and significantly accelerating reaction kinetics [6], [7]. These systems enhance antigen-antibody interactions, enabling faster and more sensitive assays. Several microfluidic platforms have been developed, including lab-on-a-chip systems, droplet-based microfluidics, and paper-based devices [7], [8]. Among these, microcapillary film (MCF) technology has gained attention as a versatile platform composed of parallel microcapillaries that provide a high surface area for antibody immobilization and support efficient, high-throughput immunoassays under controlled flow conditions [9].

Immunoassays rely on the highly specific interaction between antibodies and antigens, enabling the detection of target proteins at very low concentrations [10]. Immunoassays can be broadly classified as homogeneous or heterogeneous depending on whether separation steps are required during the analytical process. In homogeneous immunoassays, detection occurs directly in solution without the need to remove unbound reagents. In contrast, heterogeneous immunoassays involve separation or washing steps to remove unbound molecules and are frequently implemented on solid-phase platforms [10] with immobilized biomolecules. The immobilization of these molecules alters their structure and therefore their activity and formation of antibody-antigen complex, which directly influences the performance of the immunoassays.

Immunoassay performance is defined by key parameters such as the limit of detection (LOD), dynamic range, precision, accuracy, and sensitivity, reflecting the assay's ability to detect low concentrations, quantify across a range, and produce reliable and responsive measurements [11].

Biological matrix components, including heterophilic antibodies, plasma proteins, lipids, and small molecules like biotin, can interfere with immunoassay performance by disrupting antigen-antibody interactions or generating false signals [12], [13], [14]. Conventional mitigation strategies (sample dilution, blocking agents, centrifugation, filtration) are effective but add complexity, time, and equipment requirements, while dilution can compromise sensitivity [15], [16]. These challenges are amplified in miniaturized rapid diagnostic systems, where short incubation times make assays more vulnerable to even low-affinity interferents that would be inconsequential in laboratory settings. Although plasma separation approaches partially address this, interferents persist in plasma itself, underscoring the importance of developing and validating miniaturized assays directly in complex biological matrices to ensure clinical translation [15], [17].

The MCF is a mass-manufactured fluoropolymer microfluidic platform based on Teflon fluorinated ethylene propylene (FEP) that has demonstrated unmatched sensitivity for optical quantitation of clinically relevant biomarkers, relying on passive adsorption of diagnostic IgG antibodies onto its hydrophobic inner surface [18], [19] (**Figure 1A**). In addition to high optical transparency, this specialized architecture allows for efficient multiplexing integrated with portable smartphone imaging detection systems

[20] **(Figure 1B)**. Despite the counterintuitive "nonstick" reputation of fluoropolymers, passive adsorption onto FEP microcapillaries has proven rapid (equilibrium within 5–10 min), robust, and highly irreversible, forming a bioactive antibody monolayer insensitive to temperature and ionic strength [16]. Experimental studies have established that antibody surface coverage governs both the orientation and antigen-binding capacity of adsorbed IgG: at low densities antibodies adopt a flat-on configuration with Fab fragments in contact with the surface, transitioning to end-on orientations at intermediate coverages that are optimal for immunoassay performance [21], [22], [23]. **(Figure 1C)**. Critically, antibody surface coverage also determines resistance to biological matrix interference, a key challenge in rapid miniaturized diagnostics, as low-affinity serum components such as heterophilic antibodies compete with high-affinity antigens for immobilized antibody binding sites, and only a well-formed, optimally dense monolayer with properly oriented Fab regions can outcompete these interferents [16]. However, the molecular-level mechanisms governing these adsorption phenomena on FEP, including the specific residue-surface interactions, conformational dynamics, and surface coverage-dependent orientational transitions, remain poorly understood, limiting rational design of MCF-based assays. Molecular dynamics (MD) simulation offers a powerful route to resolve these mechanisms at atomic resolution, and coarse-grained approaches in particular enable exploration of the relevant length and time scales for protein adsorption on polymer surfaces.

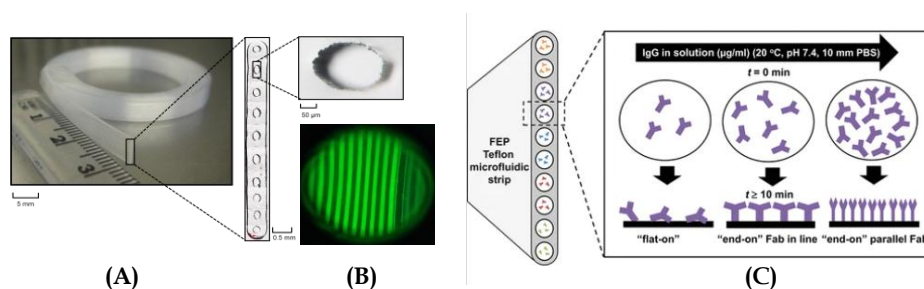


Figure 1: Overview of the Microcapillary Film (MCF) diagnostic platform and passive antibody adsorption mechanisms. **(A)** Cross-sectional architecture and parallel microcapillary structure of the fluoropolymer FEP film. **(B)** Diagnostic readout profile using fluorophore targets monitored via portable smartphone imaging frameworks. **(C)** Schematic representation of density-dependent IgG adsorption at the hydrophobic FEP interface, illustrating the conformational transition from an inactive flat-on state to a highly bioactive, vertically oriented end-on monolayer

MD simulations have become an important tool for studying the adsorption of biomolecules onto solid surfaces at atomic resolution, enabling mechanistic insights that are inaccessible to experimental methods alone. All-atom simulations using established force fields such as OPLS-AA/L and CHARMM36m, implemented in software packages such as GROMACS, have been widely used to characterize the conformational dynamics, solvent interactions, and surface binding behavior of antibody fragments [24], [25]. Specifically, previous all-atom molecular dynamics studies and experimental characterizations have focused on how individual domains, such as the Fab and Fc fragments, interact with hydrophobic solid substrates [22]. Simulations utilizing advanced atomistic force fields have revealed that the initial anchoring of these fragments onto hydrophobic polymers is driven by patches of solvent-exposed non-polar residues, where aromatic amino acids undergo localized conformational adaptations to maximize van der Waals and dispersive contacts with the solid surface [26]. Furthermore, atomistic trajectories have shown that the dynamic behaviour of the protein backbone shifts depending on the surface properties, frequently restricting the antigen-binding loops (CDRs) into specific orientations that dictate the residual bioactivity of the monolayer [22].

However, these high-fidelity simulations highlight a critical thermodynamic challenge: the passive adsorption process onto solid hydrophobic surfaces is heavily governed by the displacement of the highly structured hydration layer at the solid-liquid interface [27]. Accurately tracking this water expulsion and the subsequent long-term structural relaxation of a full-length IgG requires simulation timescales that are currently inaccessible to traditional all-atom frameworks. Because explicitly modelling every atom of a complete antibody in a fully solvated system creates an immense computational bottleneck, atomistic trajectories remain restricted to small isolated domains and limited to the tens or hundreds of nanoseconds regime, thereby establishing a clear necessity for systematically parameterized coarse-grained approaches.

Studies of protein-surface interactions have revealed that water displacement plays a central thermodynamic role: when molecules adsorb onto hydrophobic surfaces, entropic gain from releasing ordered surface water molecules is a primary driving force, and MD simulations have been used to quantify the number of water molecules displaced from protein hydration shells and the resulting changes in protein flexibility [28]. Competing

interactions between the protein, solvent molecules, and surface-bound species further modulate adsorption behavior, for instance, multivalent buffer components such as citrate can form stable bridging hydrogen bonds to protein surface residues, and their displacement by other molecules significantly alters protein dynamics [29], [30]. The orientation and conformational stability of adsorbed antibodies is also pH-dependent, as protonation states of ionizable residues alter charge distributions and inter-domain flexibility, affecting both the elbow angle and solvent-accessible surface area of aggregation-prone regions [31]. Coarse-grained approaches extend these capabilities to larger systems and longer timescales, making them particularly suited for studying adsorption at the scale of full antibody monolayers on microfluidic surfaces [17]. However, the majority of existing MD studies focus on antibody stability in bulk solution or simple buffer environments, and do not address adsorption onto fluoropolymer surfaces such as Teflon FEP or the interplay between surface chemistry, antibody orientation, and the matrix interference phenomena relevant to rapid miniaturized diagnostics [32].

Project Objectives

The central problem of this project is the limited mechanistic understanding of how antibodies adsorb onto hydrophobic fluoropolymer surfaces such as Teflon FEP, which forms the inner wall of microcapillary film (MCF) rapid diagnostic platforms. Antibody surface coverage and orientation on FEP directly determine both antigen-binding capacity and resistance to matrix interference from low-affinity serum components. Molecular dynamics simulation offers a route to atomic-level insight into these interactions but requires that the computational model first accurately reproduces the fundamental surface properties of FEP before more complex phenomena such as protein adsorption can be meaningfully simulated.

The main objective is to validate a molecular dynamics model of the Teflon FEP surface in GROMACS by reproducing the experimentally known water contact angle, and to characterise how this surface property varies with temperature and ionic strength, establishing the computational foundation necessary for future simulation of antibody adsorption onto MCF microcapillary surfaces.

The project follows a computational approach implemented in GROMACS. Contact angle simulations were performed for the pure water on

FEP system across varying operational temperatures. Results were compared against experimentally established values, including the known FEP water contact angle which typically ranges between 108° and 120° under ambient conditions [18] and the temperature insensitivity of antibody adsorption reported in the literature. This work constitutes the essential force field validation step that must precede any reliable simulation of antibody-FEP interactions, and the findings provide a characterised model of the FEP surface ready for use in future antibody adsorption studies.

To systematically quantify this validation profile, the simulated contact angle is evaluated using a manual geometric method in VMD applied to the equilibrated trajectories. Once thermodynamic equilibrium is verified via the stabilization of the system's short-range interaction energies, configurations are analyzed between 600 ps and 1000 ps (steps in GROMACS). Within this steady-state window, representative frames are extracted at regular sampling time steps of 20 ps intervals. By capturing a perpendicular side view of the substrate, the droplet's boundary is approximated as a circle to draw a tangent line at the triple-line contact point.

Methodology & Computational Tools

To address the core objectives proposed in this work and decode the interfacial mechanisms driving antibody adsorption onto FEP-Teflon, a computational approach based on coarse-grained molecular dynamics simulations was adopted. This methodology allows for detailed investigation of biomolecular interactions occurring at complex solid-liquid biointerfaces while maintaining computational costs strictly compatible with the extended timescales and large system sizes relevant for microfluidic diagnostic platforms.

The simulations were performed using the GROMACS molecular dynamics package and the Statistical Associating Fluid Theory (SAFT- γ) coarse-grained force field. Unlike traditional atomistic approaches, where every single atom and its respective fast-vibrating bonds are explicitly represented, coarse-grained models systematically group several heavy atoms into larger, discrete interaction sites, commonly referred to as beads. This structural simplification substantially reduces the total number of particles and degrees of freedom within the simulation box. Consequently, it enables the simulation of larger multi-component systems over longer timescales, moving

from the picosecond architecture up to the microsecond regime, while successfully preserving the essential thermodynamic, structural, and interfacial properties of the materials involved (**Figure 2**).

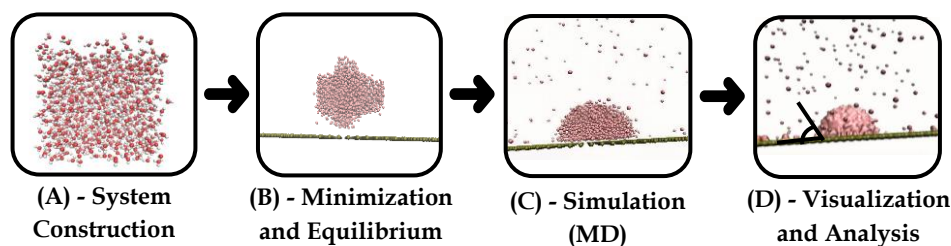


Figure 2: Computational workflow for the calculation of the water contact angle on the FEP-Teflon surface. The process initiates with (A) the system setup and execution of the coarse-grained molecular dynamics production runs; followed by (B) trajectory structural equilibration checking, separating the transient phase from the steady-state window; (C) extraction of the fluid density profiles and droplet geometric parameters using VMD analysis; and concludes with (D) the statistical computation of the final time-averaged contact angles (0-40 °C)

Varying temperature in the contact angle simulations adds a further layer of validation. In the literature, the water contact angle of fluoropolymers like FEP exhibits a minor, linear decrease with increasing temperature, typically at a rate of approximately $-0.1^{\circ}/\text{C}$ across moderate ranges [33], [34]. Only once the simulated FEP surface correctly reproduces both the magnitude and temperature dependence of the water contact angle can the model be trusted as a reliable foundation for simulating the molecular events governing antibody adsorption, orientation, and surface coverage that ultimately determine MCF diagnostic performance.

Water-Teflon Model

A simplified, multi-phase system composed of a solid fluoropolymer slab and a micro-confined water droplet was constructed to evaluate the wettability and surface forces of the fluoropolymer substrate. Teflon was selected as the target substrate because fluorinated ethylene propylene (FEP), modelled here using perfluorohexane (PFH) chains, is used in microcapillary film (MCF) technologies employed in rapid quantitative diagnostic devices due to its optical clarity and chemical inertness.

The simulation box consisted of a flat fluoropolymer surface composed of a dense matrix of perfluorohexane (PFH) chains, positioned at the bottom of the system to act as the capillary wall boundary, with a pre-equilibrated water droplet placed directly above it. This PFH model accurately reproduces the characteristic low surface energy and fluorocarbon-solvent interactions of commercial Teflon. The morphological evolution of the droplet shape during the production run was closely monitored over time and used to calculate the contact angle formed between the liquid phase boundary and the solid surface.

Contact angle measurements provide a direct quantitative assessment of surface wettability and constitute one of the most widely accepted parameters for validating novel or parameterized surface models against macroscopic experimental data. Literature reports contact angles between approximately 108° to 120° for pure water on clean Teflon surfaces under standard ambient conditions [35]. Therefore, reproducing values within this precise experimental range was considered an important baseline criterion for model validation, ensuring that the Mie potential governing the SAFT- γ beads correctly mirror real-world non-bonded interactions.

Simulation Protocol

The complete simulation workflow was engineered as a structured pipeline consisting of four main stages implemented in GROMACS: system construction, energy minimization, thermodynamic equilibration, and production molecular dynamics. Initially, the system geometry was generated by merging the coordinate files of the relaxed PFH substrate, where short-chain oligomers were used to successfully reproduce the density, low surface energy, and hydrophobicity of macroscopic Teflon FEP, and the uncharged solvent block. Energy minimization was subsequently performed using a steepest-descent algorithm to effectively remove initial steric clashes and unphysical atomic overlaps from manual placement.

Following minimization, a dedicated equilibration stage allowed for temperature stabilization, local density adjustments, and the physical relaxation of the water droplet at the solid interface. Once steady-state equilibrium conditions were verified through energy drift monitoring, long production runs tracked the dynamic evolution of the water-Teflon interface over time. Trajectory analysis was carried out using Visual Molecular Dynamics (VMD) software, where representative frames were extracted at

regular intervals to apply geometric fitting algorithms to the droplet profile and determine the resulting contact angles.

Results

The temporal progression of the coarse-grained molecular dynamics trajectories reveals a highly dynamic process of physical relaxation as the initial water block reorganizes over the non-polar PFH surface. During the opening phases, the water beads undergo a rapid morphological transition driven by the imbalance of cohesive bulk forces and weak adhesive interactions at the polymer interface, causing the fluid to contract into a well-defined droplet profile.

A detailed analysis showed that the system successfully overcomes these initial mechanical stresses, achieving a stable thermodynamic equilibrium starting from 600 ps. Prior to this checkpoint, the measured contact angles display high statistical noise reflecting transient fluid settling. Therefore, to eliminate transient artifacts, all data sampling was strictly restricted to the trajectory window extending from 600 ps to 1000 ps, where the droplet morphology oscillates gently around a steady-state mean (**Figure 3**).

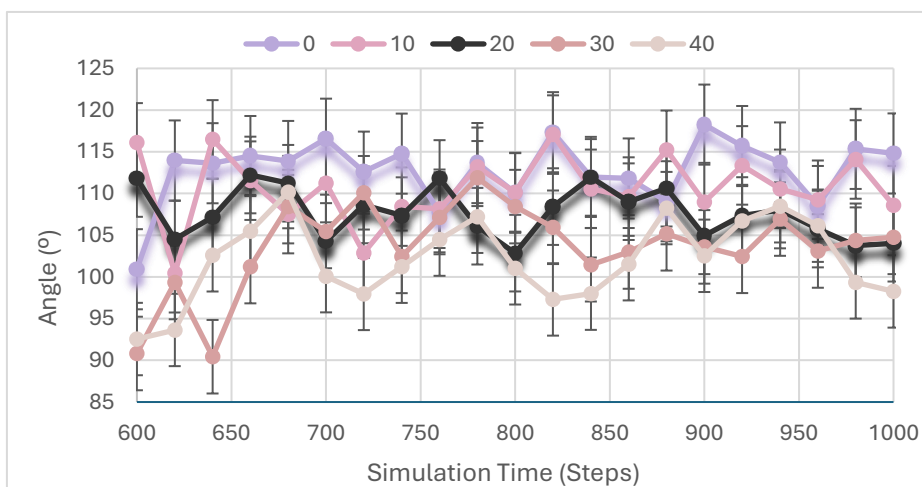


Figure 3: Dynamic evolution and stability of the simulated water contact angles on the FEP-Teflon surface as a function of simulation time (ps). The trajectories display the steady-state relaxation window from 600 ps to 1000 ps, compiled for five distinct operational temperatures: 0°C, 10°C, 20°C, 30°C and 40°C. Data points were extracted from the production runs at regular sampling intervals of 20 ps to capture local interfacial fluctuations. Error bars indicate the statistical standard deviation derived from the droplet's cylindrical symmetry analysis at each individual time point

Quantifying the steady-state contact angle as a function of temperature uncovered a highly consistent, monotonic downward trend as the environmental temperature was elevated from 0 °C to 40 °C. The average contact angle values computed across all simulated conditions fell strictly within the 108° to 120° window, matching experimental literature values for clean, commercial-grade FEP-Teflon surfaces with high fidelity (**Figure 4**). The narrow standard deviations recorded across the entire sampling timeline further confirm the structural stability of the model, proving it is completely free from anomalous density spikes or local phase separation.

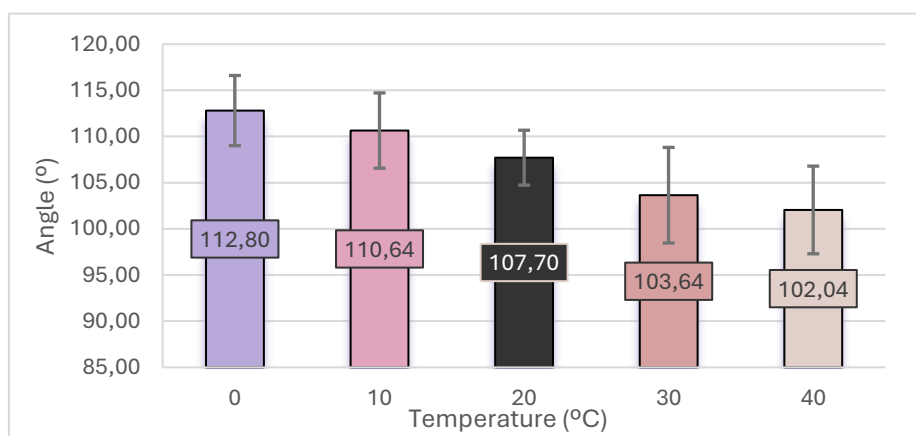


Figure 4: Time-averaged water contact angles on the FEP-Teflon surface as a function of system temperature (0°C to 40°C). Bars represent the mean values computed across the stable, steady-state production trajectory window (600-1000 ps), explicitly illustrating the monotonic downward trend in surface wettability with increasing thermal energy. Error bars denote the global standard deviation associated with the dynamic fluctuations of the droplet profile over time

Discussion

The observed negative correlation between the contact angle and temperature can be explained by standard interfacial thermodynamics. As thermal energy is introduced, the kinetic energy of individual SAFT- γ water beads increases, disrupting the hydrogen-bonding network that holds the bulk liquid together and decreasing cohesive energy and surface tension. Concurrently, weak dispersive forces between water beads and fluorocarbon chains become relatively more influential, allowing the droplet to overcome internal tension and spread further across the PFH substrate, yielding a progressively lower contact angle.

While the baseline hydrophobicity of the polymer remains high across the thermal spectrum, these subtle shifts in wettability carry major operational implications for quantitative microfluidic diagnostics. In rapid point-of-care assays, immunochemical binding reactions occur within highly restricted volumes and short windows, meaning the system frequently functions far from true thermodynamic equilibrium. Small variations in ambient temperatures can subtly alter the balance of local interfacial forces during the critical surface-coating phase.

Under higher ambient temperatures, the slightly increased surface wettability changes the structural arrangement and packing density of the pre-adsorbed IgG capture antibodies. As the surface becomes marginally less repellent, the conformational freedom of the protein backbone shifts, potentially leading to unpredictable orientations or partial crowding of the active regions. This structural variation directly dictates the spatial accessibility of the immobilized binding sites to the incoming target analyte, risking assay reproducibility and worsening matrix interference caused by low-affinity, highly abundant components in real patient serum, such as heterophilic antibodies.

This temperature-dependent behaviour observed in our simulations, is consistent with literature studies that report contact angles of water in fluoropolymer surfaces. For the pure water–Teflon interface, the water contact angle decreased with increasing temperature, consistent with experimental observations on hydrophobic fluoropolymer surfaces and values reported in the MD simulation literature [33], [34], [36]. This trend reflects the increased kinetic energy of water beads at higher temperatures, reducing the orientational order of the water–surface interface and lowering the free energy penalty for water to wet the hydrophobic surface. Reproducing this trend successfully validates the force field parameterization for the FEP–water interface before introducing biological molecules.

In contrast, no significant temperature dependence on antibody adsorption was reported in the more complex system comprising antibodies, phosphate-buffered saline (PBS) at the FEP surface. This is highly consistent with experimental data showing that IgG adsorption onto FEP microcapillaries is largely insensitive to temperature [18] and ionic strength [19] across the physiologically relevant range (4–37°C). The absence of a thermal effect in these adsorption simulations reflects the fact that antibody binding onto

hydrophobic surfaces is primarily an entropically driven process. The release of ordered water molecules from both the protein surface and the FEP interface provides the dominant thermodynamic driving force, an entropic contribution that changes relatively little across the temperatures studied. Furthermore, the additional complexity of the buffered system, including ion-surface interactions and protein-solvent competition, may further mask the subtle temperature-dependent trends that are apparent in the simpler water-only case.

Crucially, the temperature-dependent wettability of the pure fluid does not contradict the experimental temperature-independence of antibody adsorption onto FEP. It simply demonstrates that the overall process reflects the dominant entropic contribution of water release upon protein-surface contact rather than changes in surface tension alone. The contact angle results therefore confirm that the simulated FEP surface correctly reproduces the hydrophobic driving force central to antibody adsorption, providing a validated computational model ready for future coarse-grained simulations of IgG adsorption, orientation, and surface coverage on MCF microcapillary surfaces.

Conclusion

In this study, a coarse-grained molecular dynamics protocol using the SAFT- γ force field was successfully established and validated to characterize the water-Teflon interface. The model accurately replicated the intrinsic hydrophobicity of the fluorinated substrate and its thermodynamic sensitivity to temperature, yielding contact angle values in strict agreement with literature benchmarks. These findings demonstrate that temperature is an operational variable that modulates surface wettability, potentially altering the structural arrangement and functional accessibility of capture biomolecules immobilized inside diagnostic microcapillaries. Ultimately, this successful validation completes the first major milestone of this research, delivering a stable, calibrated computational foundation for upcoming project stages, which will introduce explicit electrolytes to mimic human serum, understand immobilised antibody orientation correlation with antibody surface density and unravel the mechanisms of competitive matrix interference driven by heterophilic antibodies.

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